THE COMPLETE MATHEMATICS ROADMAP

Counting se Research tak — Harvard · MIT · Oxford · Cambridge Level

Yeh PDF mathematics ke **har ek topic ki ordered list** hai — bilkul shuruaat (3-saal ki ginti) se lekar pure research-frontier mathematics tak. Koi bhi topic nahi choda gaya. Har topic is sequence mein hai ke ek topic dusre ki buniyaad banata hai. Is list ko follow karo — Harvard, MIT, Oxford, Cambridge ke darwaze khulenge.

LEVEL	RANGE	TARGET
Level 1–2	Nursery–Class 2	Basic Numeracy
Level 3–5	Class 3–6	Arithmetic Mastery
Level 6–8	Class 7–10	Algebra & Geometry
Level 9–11	Class 11–12 / A-Level	Pre-University
Level 12–14	BSc Year 1–3	Undergraduate
Level 15–16	MSc	Graduate
Level 17	PhD / Research	Frontier Mathematics

CHAPTER 1 — FOUNDATION NUMERACY (Level 1 | Age 3–6)

1.1 Numbers aur Counting

1.1.1 Rote counting: 1 to 10 (bolna)
1.1.2 Rote counting: 1 to 20
1.1.3 Number recognition: symbols 0–9 pehchanna
1.1.4 One-to-one correspondence — ek cheez, ek number
1.1.5 Cardinality — last number = total count
1.1.6 Counting objects in a set (1–10)
1.1.7 Zero ka concept
1.1.8 Counting forward and backward (1–20)
1.1.9 Number before / after
1.1.10 Ordinal numbers: 1st, 2nd, 3rd … 10th

1.2 Comparison aur Ordering

1.2.1 More / Less / Equal
1.2.2 Bigger / Smaller (size)
1.2.3 Ordering numbers 1–10 on a line
1.2.4 Heaviest / Lightest (non-standard measure)
1.2.5 Longest / Shortest (direct comparison)

1.3 Basic Shapes

1.3.1 2D shapes: Circle, Square, Triangle, Rectangle
1.3.2 3D shapes: Sphere, Cube, Cylinder, Cone
1.3.3 Sorting shapes by attribute
1.3.4 Positions: above, below, beside, inside, outside

1.4 Patterns

1.4.1 Recognising repeating patterns: AB, AAB, ABC
1.4.2 Copying and extending patterns
1.4.3 Creating own patterns

1.5 Basic Data

1.5.1 Sorting objects into groups
1.5.2 Simple pictographs
1.5.3 Yes/No charts

CHAPTER 2 — PRIMARY ARITHMETIC (Level 2 | Age 6–9)

2.1 Number System

2.1.1 Numbers 1–100: read, write, count
2.1.2 Place value: Tens and Ones
2.1.3 Numbers 1–1000: Hundreds, Tens, Ones
2.1.4 Expanded form: 345 = 300+40+5
2.1.5 Comparing numbers: >, <, =
2.1.6 Ordering numbers on a number line
2.1.7 Even and Odd numbers
2.1.8 Skip counting: 2s, 3s, 4s, 5s, 10s
2.1.9 Roman numerals: I to XX
2.1.10 Numbers to 10,000

2.2 Addition

2.2.1 Addition facts to 10 (memorisation)
2.2.2 Addition facts to 20
2.2.3 Adding 2-digit numbers without carrying
2.2.4 Adding 2-digit numbers with carrying (regrouping)
2.2.5 Adding 3-digit numbers
2.2.6 Column addition (multiple numbers)
2.2.7 Adding 4-digit numbers
2.2.8 Mental addition strategies (make-10, doubles)
2.2.9 Word problems — addition

2.3 Subtraction

2.3.1 Subtraction facts to 10
2.3.2 Subtraction facts to 20
2.3.3 Subtracting 2-digit numbers without borrowing
2.3.4 Subtracting with borrowing (decomposition)
2.3.5 Subtracting 3-digit and 4-digit numbers
2.3.6 Checking subtraction with addition
2.3.7 Word problems — subtraction

2.4 Multiplication

2.4.1 Concept: repeated addition
2.4.2 Arrays — rows and columns
2.4.3 Times tables: 1, 2, 5, 10 (fluency)
2.4.4 Times tables: 3, 4, 6, 7, 8, 9 (fluency)
2.4.5 Times tables: 11, 12
2.4.6 Commutativity: 3×4 = 4×3
2.4.7 Multiplying by 10, 100, 1000
2.4.8 Short multiplication: 2-digit × 1-digit
2.4.9 Word problems — multiplication

2.5 Division

2.5.1 Concept: equal sharing and grouping
2.5.2 Division as inverse of multiplication
2.5.3 Division facts (linked to times tables)
2.5.4 Remainders
2.5.5 Short division (bus stop method)
2.5.6 Dividing by 10, 100
2.5.7 Word problems — division

2.6 Fractions (Introduction)

2.6.1 Unit fractions: 1/2, 1/3, 1/4, 1/5, 1/8, 1/10
2.6.2 Non-unit fractions: 2/3, 3/4, 3/5
2.6.3 Fractions of shapes (shading)
2.6.4 Fractions of quantities: 1/4 of 20
2.6.5 Comparing fractions with same denominator
2.6.6 Equivalent fractions (picture method)
2.6.7 Mixed numbers: 2½

2.7 Measurement

2.7.1 Length: cm, m, km — measure and compare
2.7.2 Weight: g, kg
2.7.3 Capacity: ml, L
2.7.4 Temperature: Celsius reading
2.7.5 Perimeter: measuring around a shape
2.7.6 Area: counting squares
2.7.7 Time: o'clock, half past, quarter past/to
2.7.8 Reading analogue and digital clocks
2.7.9 Calendar: days, weeks, months, years
2.7.10 Money: coins and notes, giving change

2.8 Geometry

2.8.1 Properties of 2D shapes: sides, angles, vertices
2.8.2 Properties of 3D shapes: faces, edges, vertices
2.8.3 Right angles — recognising
2.8.4 Horizontal, vertical, perpendicular, parallel lines
2.8.5 Lines of symmetry
2.8.6 Reflection in a mirror line
2.8.7 Compass directions: N, S, E, W
2.8.8 Co-ordinate grids (first quadrant)

2.9 Data Handling

2.9.1 Tally charts
2.9.2 Bar charts — draw and interpret
2.9.3 Pictograms
2.9.4 Frequency tables
2.9.5 Venn diagrams (two sets)

CHAPTER 3 — INTERMEDIATE ARITHMETIC (Level 3 | Age 9–11)

3.1 Advanced Number

3.1.1 Numbers to 1,000,000 — read, write, order
3.1.2 Lakhs and Crores (Indian system)
3.1.3 Millions and Billions (International system)
3.1.4 Negative numbers — concept and number line
3.1.5 Factors and Multiples
3.1.6 Factor pairs
3.1.7 Prime numbers (up to 100)
3.1.8 Composite numbers
3.1.9 Prime factorisation (factor trees)
3.1.10 HCF (Highest Common Factor)
3.1.11 LCM (Lowest Common Multiple)
3.1.12 Square numbers 1–15 squared
3.1.13 Cube numbers 1–5 cubed
3.1.14 Square roots (perfect squares)
3.1.15 Powers: base and exponent notation

3.2 Long Multiplication and Division

3.2.1 Long multiplication: 3-digit × 2-digit
3.2.2 Long multiplication: 4-digit × 2-digit
3.2.3 Grid method
3.2.4 Long division: 3-digit ÷ 1-digit
3.2.5 Long division: 3-digit ÷ 2-digit
3.2.6 Short and long division with remainders
3.2.7 Divisibility rules: 2,3,4,5,6,8,9,10,11

3.3 Fractions — Operations

3.3.1 Simplifying fractions
3.3.2 Equivalent fractions
3.3.3 Converting: improper fractions ↔ mixed numbers
3.3.4 Adding fractions — same denominator
3.3.5 Adding fractions — different denominators
3.3.6 Subtracting fractions — same denominator
3.3.7 Subtracting fractions — different denominators
3.3.8 Multiplying fractions: a/b × c/d
3.3.9 Dividing fractions: a/b ÷ c/d = a/b × d/c
3.3.10 Fractions of amounts: 3/4 of 48
3.3.11 Ordering fractions on a number line

3.4 Decimals

3.4.1 Decimal place value: tenths, hundredths, thousandths
3.4.2 Converting fractions to decimals
3.4.3 Converting decimals to fractions
3.4.4 Ordering and comparing decimals
3.4.5 Rounding decimals to 1 and 2 decimal places
3.4.6 Adding and subtracting decimals
3.4.7 Multiplying decimals by 10, 100, 1000
3.4.8 Multiplying decimals: 2.4 × 3
3.4.9 Dividing decimals: 4.8 ÷ 4
3.4.10 Recurring decimals concept: 1/3 = 0.333…

3.5 Percentages

- 3.5.1 Percent concept: per hundred
- 3.5.2 Converting: fractions ↔ decimals ↔ percentages
- 3.5.3 Finding a percentage of a quantity
- 3.5.4 Percentage increase and decrease
- 3.5.5 Expressing one number as a percent of another
- 3.5.6 Profit and Loss percentage
- 3.5.7 Discount calculations
- 3.5.8 VAT / Tax calculations

3.6 Ratio and Proportion

- 3.6.1 Ratio notation: a:b
- 3.6.2 Simplifying ratios
- 3.6.3 Dividing a quantity in a given ratio
- 3.6.4 Unitary method (one unit then scale)
- 3.6.5 Direct proportion
- 3.6.6 Inverse proportion
- 3.6.7 Map scales and similar figures

3.7 Geometry — Angles

- 3.7.1 Types of angles: acute, right, obtuse, straight, reflex, full
- 3.7.2 Measuring angles with a protractor
- 3.7.3 Drawing angles
- 3.7.4 Angles on a straight line (sum = 180)
- 3.7.5 Angles at a point (sum = 360)
- 3.7.6 Vertically opposite angles
- 3.7.7 Angles in a triangle (sum = 180)
- 3.7.8 Angles in a quadrilateral (sum = 360)
- 3.7.9 Exterior angles of polygons
- 3.7.10 Interior angles of regular polygons

3.8 Geometry — Shapes and Constructions

- 3.8.1 Types of triangles: equilateral, isosceles, scalene, right-angled
- 3.8.2 Types of quadrilaterals: square, rectangle, parallelogram, rhombus, trapezium, kite
- 3.8.3 Properties of regular polygons (pentagon to decagon)
- 3.8.4 Circle: centre, radius, diameter, chord, arc, sector, segment, tangent
- 3.8.5 Compass constructions: perpendicular bisector, angle bisector
- 3.8.6 Constructing triangles (SSS, SAS, ASA)
- 3.8.7 Loci: sets of points satisfying conditions

3.9 Area and Perimeter

- 3.9.1 Area of rectangle and square
- 3.9.2 Area of triangle
- 3.9.3 Area of parallelogram
- 3.9.4 Area of trapezium
- 3.9.5 Compound shapes — area and perimeter
- 3.9.6 Circumference of circle: $C = 2\pi r$
- 3.9.7 Area of circle: $A = \pi r^2$
- 3.9.8 Arc length and sector area

3.10 Volume and Surface Area (Introduction)

- 3.10.1 Volume of cuboid: $l \times w \times h$
- 3.10.2 Volume of cube
- 3.10.3 Surface area of cuboid

3.10.4 Surface area of cube
3.10.5 Volume concept using unit cubes

3.11 Statistics (Intermediate)

3.11.1 Mean (average)
3.11.2 Median
3.11.3 Mode
3.11.4 Range
3.11.5 Mean from a frequency table
3.11.6 Grouped data — modal class
3.11.7 Bar charts, line graphs, pie charts — draw and interpret
3.11.8 Scatter graphs — plotting points
3.11.9 Correlation — positive, negative, none
3.11.10 Line of best fit

3.12 Probability (Introduction)

3.12.1 Probability scale: 0 to 1
3.12.2 Language: impossible, unlikely, even, likely, certain
3.12.3 Theoretical probability: P(E) = favourable / total
3.12.4 Experimental probability
3.12.5 Combined events: sample space diagram
3.12.6 Mutually exclusive events: P(A or B) = P(A)+P(B)

CHAPTER 4 — PRE-ALGEBRA TO ALGEBRA (Level 4 | Age 11–13)

4.1 Algebraic Thinking

4.1.1 Variables: using letters for unknowns
4.1.2 Writing algebraic expressions
4.1.3 Substitution into expressions
4.1.4 Simplifying: collecting like terms
4.1.5 Expanding single brackets: 3(2x+1)
4.1.6 Factorising: taking out common factor
4.1.7 Index laws: a^m × a^n = a^(m+n)
4.1.8 Index laws: (a^m)^n = a^(mn)
4.1.9 Zero and negative indices
4.1.10 BODMAS/PEMDAS rule with algebra
4.1.11 Formula: writing and using

4.2 Linear Equations (One Variable)

4.2.1 Solving one-step equations: x+5=12
4.2.2 Solving two-step equations: 2x+3=11
4.2.3 Equations with brackets: 3(x+2)=15
4.2.4 Equations with unknowns on both sides
4.2.5 Forming and solving equations from word problems
4.2.6 Equations with fractions

4.3 Linear Inequalities

4.3.1 Inequality notation: <, >, ≤, ≥
4.3.2 Solving linear inequalities
4.3.3 Representing solutions on a number line
4.3.4 Integer solutions of inequalities
4.3.5 Double inequalities: 2 < 3x-1 ≤ 11

4.4 Sequences

4.4.1 Term-to-term rule
4.4.2 Position-to-term rule (nth term)
4.4.3 Linear sequences: nth term = dn + c
4.4.4 Quadratic sequences (introduction)
4.4.5 Geometric sequences: common ratio
4.4.6 Fibonacci sequence and special sequences
4.4.7 Sum of arithmetic series
4.4.8 Sum of geometric series (finite)
4.4.9 Sum to infinity of geometric series

4.5 Coordinate Geometry (2D)

4.5.1 Four quadrants — plotting coordinates
4.5.2 Gradient (slope) = rise/run
4.5.3 y-intercept
4.5.4 Equation of a straight line: y = mx + c
4.5.5 Drawing lines from equations
4.5.6 Finding equation from two points
4.5.7 Parallel and perpendicular lines
4.5.8 Midpoint of a line segment
4.5.9 Length of a line segment
4.5.10 Intersection of two lines

4.6 Graphs of Functions (Introduction)

4.6.1 Plotting graphs from tables of values
4.6.2 Graphs of $y = mx + c$
4.6.3 Graphs of quadratics: $y = x^2$
4.6.4 Graphs of cubics: $y = x^3$
4.6.5 Reciprocal graph: $y = 1/x$
4.6.6 Exponential graph: $y = 2^x$
4.6.7 Reading and interpreting real-life graphs
4.6.8 Distance-time graphs
4.6.9 Speed-time graphs and area under curve

4.7 Ratio, Rates and Proportion (Advanced)

4.7.1 Direct proportion: $y \propto x$, $y = kx$
4.7.2 Inverse proportion: $y \propto 1/x$
4.7.3 Other proportionalities: $y \propto x^2$, $y \propto \sqrt{x}$
4.7.4 Speed, distance, time problems
4.7.5 Density = mass / volume
4.7.6 Pressure = force / area
4.7.7 Currency conversion and exchange rates
4.7.8 Best value problems

4.8 Financial Mathematics

4.8.1 Simple Interest: $I = PRT/100$
4.8.2 Compound Interest: $A = P(1+r/100)^n$
4.8.3 Depreciation
4.8.4 Reverse percentage (finding original amount)
4.8.5 Repeated percentage change
4.8.6 Salary, tax, net income
4.8.7 Hire purchase and instalment plans

CHAPTER 5 — ALGEBRA II AND GEOMETRY II (Level 5 | Age 13–15)

5.1 Polynomials

5.1.1 Degree and terms of a polynomial
5.1.2 Addition and subtraction of polynomials
5.1.3 Multiplication: $(a+b)(c+d)$ expansion
5.1.4 Special products: $(a+b)^2$, $(a-b)^2$, $(a+b)(a-b)$
5.1.5 Factorising quadratics: $x^2 + bx + c$
5.1.6 Factorising: $ax^2 + bx + c$ $(a \neq 1)$
5.1.7 Difference of two squares
5.1.8 Sum and difference of cubes: $a^3 \pm b^3$
5.1.9 Polynomial long division
5.1.10 Remainder theorem
5.1.11 Factor theorem

5.2 Quadratic Equations

5.2.1 Solving by factorisation
5.2.2 Completing the square
5.2.3 Quadratic formula: $x = (-b \pm \sqrt{(b^2-4ac)}) / 2a$
5.2.4 Discriminant: b^2-4ac — nature of roots
5.2.5 Sum and product of roots (Vieta's formulas)
5.2.6 Forming quadratic from given roots
5.2.7 Quadratic inequalities
5.2.8 Equations reducible to quadratic form

5.3 Simultaneous Equations

5.3.1 Linear simultaneous: substitution method
5.3.2 Linear simultaneous: elimination method
5.3.3 Linear simultaneous: graphical method
5.3.4 One linear, one quadratic: simultaneous
5.3.5 Word problems — simultaneous equations
5.3.6 Three equations in three unknowns (introduction)

5.4 Functions

5.4.1 Function notation: $f(x)$, $g(x)$
5.4.2 Domain and Range
5.4.3 Composite functions: $fg(x)$
5.4.4 Inverse functions: $f^{\blacksquare 1}(x)$
5.4.5 Even and odd functions
5.4.6 Increasing and decreasing functions
5.4.7 Piecewise functions
5.4.8 Modulus (absolute value) function: $|x|$
5.4.9 Transformations of graphs: $f(x)+a$, $f(x+a)$, $af(x)$, $f(ax)$
5.4.10 Reflection: $y=-f(x)$, $y=f(-x)$

5.5 Indices and Surds

5.5.1 Index laws — complete set including fractional
5.5.2 Fractional indices: a^(1/n) = n-th root
5.5.3 Negative indices: a^(-n) = 1/a^n
5.5.4 Surds: definition — irrational square roots
5.5.5 Simplifying surds: $\sqrt{12} = 2\sqrt{3}$
5.5.6 Adding/subtracting surds
5.5.7 Multiplying surds: $\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$

5.5.8 Rationalising the denominator (monomial)
5.5.9 Rationalising the denominator (binomial: $a+\sqrt{b}$)

5.6 Logarithms

5.6.1 Definition: $\log_a(b) = c$ means $a^c = b$
5.6.2 Common logarithm log■■
5.6.3 Natural logarithm $\ln = \log_e$
5.6.4 Laws of logarithms: product, quotient, power
5.6.5 Change of base formula
5.6.6 Solving exponential equations using logs
5.6.7 Solving logarithmic equations
5.6.8 Graphs of logarithmic functions
5.6.9 Exponential growth and decay models

5.7 Geometry — Triangles and Proofs

5.7.1 Congruence criteria: SSS, SAS, ASA, AAS, RHS
5.7.2 Similarity criteria: AA, SAS, SSS
5.7.3 Scale factors — area and volume
5.7.4 Pythagoras theorem — proof and applications
5.7.5 Pythagoras in 3D
5.7.6 Geometric proofs using congruence

5.8 Circle Theorems

5.8.1 Angle at centre = 2 × angle at circumference
5.8.2 Angles in the same segment are equal
5.8.3 Angle in a semicircle = $90°$
5.8.4 Opposite angles of a cyclic quadrilateral sum to $180°$
5.8.5 Tangent is perpendicular to radius
5.8.6 Tangents from external point are equal
5.8.7 Alternate segment theorem
5.8.8 Two circles — intersections and common tangents

5.9 Trigonometry I

5.9.1 Right-angled triangle: SOH-CAH-TOA
5.9.2 Finding angles using inverse trig functions
5.9.3 Sine Rule: $a/\sin A = b/\sin B = c/\sin C$
5.9.4 Cosine Rule: $a^2 = b^2+c^2-2bc \cos A$
5.9.5 Area of triangle = $\frac{1}{2}ab \sin C$
5.9.6 Problems in 3D using trigonometry
5.9.7 Angles of elevation and depression
5.9.8 Bearings problems

5.10 Volume and Surface Area (Advanced)

5.10.1 Volume of prism: V = cross-section area × length
5.10.2 Volume of cylinder: $V = \pi r^2 h$
5.10.3 Volume of cone: $V = (1/3)\pi r^2 h$
5.10.4 Volume of sphere: $V = (4/3)\pi r^3$
5.10.5 Volume of pyramid: V = (1/3) × base area × height
5.10.6 Surface area of cylinder
5.10.7 Surface area of cone
5.10.8 Surface area of sphere: $SA = 4\pi r^2$
5.10.9 Frustum of a cone
5.10.10 Compound 3D shapes

5.11 Transformations

5.11.1 Translation — vector notation
5.11.2 Reflection — in given lines including y=x
5.11.3 Rotation — centre, angle, direction
5.11.4 Enlargement — positive and fractional scale factors
5.11.5 Negative scale factor enlargement
5.11.6 Combined transformations
5.11.7 Inverse transformations

5.12 Vectors (Introduction)

5.12.1 Vector notation: column vectors, bold, underline
5.12.2 Adding and subtracting vectors
5.12.3 Scalar multiplication of vectors
5.12.4 Magnitude of a vector: |v|
5.12.5 Position vectors
5.12.6 Geometric proofs using vectors
5.12.7 Parallel vectors and collinearity

5.13 Statistics (Advanced)

5.13.1 Cumulative frequency graphs
5.13.2 Box-and-whisker plots
5.13.3 Histograms with unequal class widths
5.13.4 Standard deviation (data set)
5.13.5 Variance
5.13.6 Interquartile range
5.13.7 Outliers
5.13.8 Comparing distributions

5.14 Probability (Intermediate)

5.14.1 Independent events: P(A and B) = P(A) × P(B)
5.14.2 Conditional probability: P(A|B)
5.14.3 Tree diagrams
5.14.4 Venn diagrams — two and three sets
5.14.5 Addition law: $P(A\cup B) = P(A)+P(B)-P(A\cap B)$
5.14.6 Complementary events: P(A') = 1-P(A)
5.14.7 Relative frequency and simulation

CHAPTER 6 — ADVANCED SECONDARY / A-LEVEL (Level 6 | Age 15–18)

6.1 Pure Mathematics — Algebra and Calculus Preparation

6.1.1 Partial fractions: distinct linear, repeated, quadratic denominator
6.1.2 Proof by deduction, exhaustion, contradiction, counter-example
6.1.3 Proof by mathematical induction
6.1.4 Binomial theorem: (a+b)^n — integer n
6.1.5 Binomial expansion: (1+x)^n — fractional n, convergence
6.1.6 Arithmetic and Geometric series — sigma notation
6.1.7 Maclaurin series (first encounter)
6.1.8 Parametric equations — curves
6.1.9 Implicit differentiation
6.1.10 Rational functions — asymptotes, sketching
6.1.11 Inequalities involving rational expressions

6.2 Trigonometry II

6.2.1 Radians: definition, conversion
6.2.2 Arc length: $s = r\theta$
6.2.3 Sector area: $A = \frac{1}{2}r^2\theta$
6.2.4 Unit circle definition of sin, cos, tan
6.2.5 Exact values: sin 30°,45°,60° etc.
6.2.6 Graphs of sin x, cos x, tan x — features
6.2.7 Amplitude, period, phase shift, vertical shift
6.2.8 Transformations of trig graphs
6.2.9 Pythagorean identity: $\sin^2 x + \cos^2 x = 1$
6.2.10 Other identities: $1+\tan^2 x = \sec^2 x$; $1+\cot^2 x = \mathrm{cosec}^2 x$
6.2.11 Double angle formulae: sin2x, cos2x, tan2x
6.2.12 Addition formulae: sin(A±B), cos(A±B), tan(A±B)
6.2.13 Half angle substitutions
6.2.14 Product-to-sum and sum-to-product formulae
6.2.15 $R \sin(x + \alpha)$ form — harmonic form
6.2.16 Solving trigonometric equations in given intervals
6.2.17 Inverse trig functions: arcsin, arccos, arctan — domains
6.2.18 Reciprocal trig: secant, cosecant, cotangent

6.3 Calculus I — Differentiation

6.3.1 Limits concept: lim as $h \to 0$
6.3.2 Derivative from first principles
6.3.3 Differentiation of x^n
6.3.4 Differentiation of polynomial functions
6.3.5 Differentiation of sin x, cos x, tan x
6.3.6 Differentiation of e^x and a^x
6.3.7 Differentiation of ln x and log_a(x)
6.3.8 Chain rule: d/dx[f(g(x))]
6.3.9 Product rule: (uv)' = u'v + uv'
6.3.10 Quotient rule: $(u/v)' = (u'v - uv') / v^2$
6.3.11 Implicit differentiation
6.3.12 Parametric differentiation: dy/dx = (dy/dt)/(dx/dt)
6.3.13 Higher order derivatives: d^2y/dx^2
6.3.14 Gradient, tangent and normal to a curve
6.3.15 Increasing / decreasing functions
6.3.16 Stationary points — maxima, minima, inflection

6.3.17 Second derivative test
6.3.18 Optimisation (applied max/min problems)
6.3.19 Rates of change — connected rates
6.3.20 Small increments and approximation: $\delta y \approx (dy/dx)\,\delta x$

6.4 Calculus II — Integration

6.4.1 Integration as anti-differentiation
6.4.2 Indefinite integral of x^n
6.4.3 Integration of $\sin x$, $\cos x$, $\sec^2 x$
6.4.4 Integration of e^x and $1/x$
6.4.5 Integration of $1/(ax+b)$
6.4.6 Integration by inspection (reverse chain rule)
6.4.7 Definite integral — limits, numerical value
6.4.8 Area under a curve (definite integral)
6.4.9 Area between two curves
6.4.10 Area under parametric curve
6.4.11 Trapezium rule (numerical integration)
6.4.12 Simpson's rule
6.4.13 Integration by substitution
6.4.14 Integration by parts: $\int u\,dv = uv - \int v\,du$
6.4.15 Integration using partial fractions
6.4.16 Volumes of revolution about x-axis and y-axis
6.4.17 Mean value of a function

6.5 Differential Equations (First Introduction)

6.5.1 Separable variables: $dy/dx = f(x)g(y)$
6.5.2 Solving and applying general and particular solutions
6.5.3 Exponential growth/decay model: $dN/dt = kN$
6.5.4 Simple harmonic motion model
6.5.5 Newton's law of cooling model

6.6 Complex Numbers

6.6.1 Definition: $i = \sqrt{(-1)}$, $i^2 = -1$
6.6.2 Arithmetic of complex numbers: add, subtract, multiply
6.6.3 Complex conjugate
6.6.4 Modulus and argument: $|z|$, $\arg(z)$
6.6.5 Argand diagram
6.6.6 Polar / modulus-argument form: $z = r(\cos\theta + i\sin\theta)$
6.6.7 Multiplication and division in polar form
6.6.8 De Moivre's Theorem: $(\cos\theta+i\sin\theta)^n = \cos n\theta+i\sin n\theta$
6.6.9 nth roots of complex numbers
6.6.10 Euler's formula: $e^{i\theta} = \cos\theta + i\sin\theta$
6.6.11 Loci in the Argand diagram

6.7 Matrices (Introduction to Linear Algebra)

6.7.1 Matrix notation and terminology
6.7.2 Addition and subtraction of matrices
6.7.3 Scalar multiplication
6.7.4 Matrix multiplication
6.7.5 Determinant of 2×2 matrix
6.7.6 Inverse of 2×2 matrix
6.7.7 Solving simultaneous equations using matrices
6.7.8 Determinant of 3×3 matrix (cofactor expansion)
6.7.9 Inverse of 3×3 matrix

6.7.10 Geometric transformations via matrices (rotation, reflection, enlargement)
6.7.11 Composition of transformations
6.7.12 Singular matrices and non-invertibility

6.8 Vectors (3D)

6.8.1 3D vectors in component form
6.8.2 Magnitude of 3D vector
6.8.3 Unit vectors: i, j, k
6.8.4 Dot product (scalar product): $a \cdot b = |a||b|\cos \theta$
6.8.5 Perpendicularity via dot product
6.8.6 Cross product (vector product): $a \times b = |a||b|\sin \theta\ n$
6.8.7 Equation of a line in 3D: $r = a + tb$
6.8.8 Angle between two lines
6.8.9 Intersection of lines
6.8.10 Skew lines
6.8.11 Equation of a plane: $r \cdot n = d$
6.8.12 Angle between line and plane
6.8.13 Angle between two planes
6.8.14 Intersection of planes
6.8.15 Distance from point to plane

6.9 Probability and Statistics (A-Level)

6.9.1 Permutations: nPr
6.9.2 Combinations: nCr
6.9.3 Probability with combinations
6.9.4 Discrete random variables: $P(X=x)$
6.9.5 Expectation $E(X)$ and Variance $Var(X)$
6.9.6 Binomial distribution: $X \sim B(n,p)$
6.9.7 Poisson distribution: $X \sim Po(\lambda)$
6.9.8 Normal distribution: $X \sim N(\mu, \sigma^2)$
6.9.9 Standard Normal $Z \sim N(0,1)$
6.9.10 Normal approximation to binomial
6.9.11 Hypothesis testing: one-tailed and two-tailed
6.9.12 Critical region and significance level
6.9.13 Unbiased estimators of population mean and variance
6.9.14 Confidence intervals
6.9.15 Product moment correlation coefficient (PMCC)
6.9.16 Spearman's rank correlation
6.9.17 Chi-squared test for association

6.10 Numerical Methods

6.10.1 Interval bisection (bisection method)
6.10.2 Linear interpolation
6.10.3 Newton-Raphson method
6.10.4 Fixed-point iteration
6.10.5 Step-by-step numerical solution of ODEs (Euler's method)

CHAPTER 7 — UNDERGRADUATE YEAR 1 (Level 7 | BSc First Year)

7.1 Logic and Proof

7.1.1 Propositions, connectives: AND, OR, NOT, XOR
7.1.2 Implication, bi-conditional
7.1.3 Truth tables
7.1.4 Tautologies and contradictions
7.1.5 Predicate logic: quantifiers ∀, ∃
7.1.6 Negation of quantified statements
7.1.7 Proof methods: direct, contrapositive, contradiction
7.1.8 Proof by mathematical induction (strong form)
7.1.9 Well-ordering principle
7.1.10 Existence and uniqueness proofs

7.2 Set Theory

7.2.1 Sets, subsets, power set
7.2.2 Set operations: union, intersection, complement, difference
7.2.3 De Morgan's laws
7.2.4 Cartesian product
7.2.5 Relations: reflexive, symmetric, transitive
7.2.6 Equivalence relations and equivalence classes
7.2.7 Partial orders
7.2.8 Functions: injective, surjective, bijective
7.2.9 Composition and inverse of functions
7.2.10 Countable and uncountable sets (Cantor)
7.2.11 Cardinality: $|A|$, Cantor-Bernstein theorem
7.2.12 Axiom of Choice (informal)

7.3 Number Theory

7.3.1 Divisibility and the Division Algorithm
7.3.2 Euclidean Algorithm for GCD
7.3.3 Extended Euclidean Algorithm
7.3.4 Bezout's Identity
7.3.5 Fundamental Theorem of Arithmetic (proof)
7.3.6 Infinitely many primes (Euclid's proof)
7.3.7 Congruences: $a \equiv b \pmod{n}$
7.3.8 Arithmetic of congruences
7.3.9 Chinese Remainder Theorem
7.3.10 Fermat's Little Theorem
7.3.11 Euler's theorem and Euler's totient function $\phi(n)$
7.3.12 Wilson's theorem
7.3.13 Quadratic residues (introduction)
7.3.14 RSA encryption (application)

7.4 Linear Algebra I

7.4.1 Systems of linear equations
7.4.2 Row reduction and echelon form
7.4.3 Reduced row echelon form (RREF)
7.4.4 Existence and uniqueness of solutions
7.4.5 Matrices as linear transformations
7.4.6 Matrix operations — full treatment
7.4.7 Transpose, symmetric, skew-symmetric
7.4.8 Determinant — properties, cofactor expansion

7.4.9 Cramer's rule
7.4.10 Vector spaces — axioms
7.4.11 Subspaces
7.4.12 Linear independence
7.4.13 Span and basis
7.4.14 Dimension theorem
7.4.15 Row space, column space, null space
7.4.16 Rank-nullity theorem
7.4.17 Eigenvalues and eigenvectors
7.4.18 Characteristic polynomial
7.4.19 Diagonalisation
7.4.20 Cayley-Hamilton theorem

7.5 Calculus III — Multivariable

7.5.1 Functions of several variables
7.5.2 Partial derivatives
7.5.3 Higher-order partial derivatives
7.5.4 Chain rule for several variables
7.5.5 Gradient vector ∇f
7.5.6 Directional derivatives
7.5.7 Tangent planes and linear approximation
7.5.8 Critical points of $f(x,y)$ — classification
7.5.9 Lagrange multipliers for constrained optimisation
7.5.10 Double integrals over rectangles and general regions
7.5.11 Changing order of integration
7.5.12 Polar coordinates and polar integrals
7.5.13 Triple integrals
7.5.14 Cylindrical and spherical coordinates
7.5.15 Jacobian and change of variables
7.5.16 Line integrals (scalar and vector fields)
7.5.17 Green's theorem
7.5.18 Divergence theorem (Gauss's theorem)
7.5.19 Stokes' theorem
7.5.20 Conservative fields and potential functions

7.6 Real Analysis I

7.6.1 Real number system — axioms, completeness
7.6.2 Supremum and infimum
7.6.3 Archimedean property
7.6.4 Sequences — convergence, epsilon-N definition
7.6.5 Limit theorems for sequences
7.6.6 Monotone convergence theorem
7.6.7 Cauchy sequences
7.6.8 Bolzano-Weierstrass theorem
7.6.9 Limits of functions — epsilon-delta definition
7.6.10 Continuity — formal definition
7.6.11 Extreme value theorem
7.6.12 Intermediate value theorem
7.6.13 Uniform continuity
7.6.14 Differentiability — formal definition
7.6.15 Mean Value Theorem (MVT)
7.6.16 L'Hopital's rule
7.6.17 Taylor's theorem with remainder

- 7.6.18 Power series — radius of convergence
- 7.6.19 Riemann integration — formal definition
- 7.6.20 Fundamental Theorem of Calculus (rigorous proof)
- 7.6.21 Improper integrals

7.7 Ordinary Differential Equations (Full Course)

- 7.7.1 Classification: order, linearity, homogeneity
- 7.7.2 Separable equations
- 7.7.3 Linear first-order: integrating factor
- 7.7.4 Exact equations
- 7.7.5 Bernoulli equations
- 7.7.6 Second-order linear homogeneous: constant coefficients
- 7.7.7 Characteristic equation — distinct, repeated, complex roots
- 7.7.8 Second-order linear non-homogeneous: method of undetermined coefficients
- 7.7.9 Variation of parameters
- 7.7.10 Cauchy-Euler equation
- 7.7.11 Power series solutions
- 7.7.12 Frobenius method
- 7.7.13 Bessel's equation and Bessel functions
- 7.7.14 Legendre's equation and Legendre polynomials
- 7.7.15 Systems of ODEs — matrix method
- 7.7.16 Phase plane analysis
- 7.7.17 Stability of equilibria
- 7.7.18 Laplace transform — definition and properties
- 7.7.19 Solving ODEs with Laplace transforms
- 7.7.20 Convolution theorem

7.8 Abstract Algebra I — Group Theory

- 7.8.1 Binary operations
- 7.8.2 Groups — definition and axioms
- 7.8.3 Examples: Z, Q, R, Zn, Sn, Dn
- 7.8.4 Order of a group and order of an element
- 7.8.5 Subgroups — definition and subgroup test
- 7.8.6 Cyclic groups and generators
- 7.8.7 Cosets — left and right
- 7.8.8 Lagrange's theorem
- 7.8.9 Normal subgroups
- 7.8.10 Quotient groups G/N
- 7.8.11 Group homomorphisms
- 7.8.12 Isomorphism theorems: First, Second, Third
- 7.8.13 Permutation groups: Sn, cycle notation
- 7.8.14 Even and odd permutations, alternating group An
- 7.8.15 Cayley's theorem
- 7.8.16 Group actions — orbits and stabilisers
- 7.8.17 Orbit-stabiliser theorem
- 7.8.18 Burnside's lemma
- 7.8.19 Sylow theorems (statement and basic applications)

7.9 Discrete Mathematics

- 7.9.1 Graph theory: vertices, edges, paths, cycles
- 7.9.2 Connectivity, trees, spanning trees
- 7.9.3 Bipartite graphs
- 7.9.4 Graph coloring — chromatic number

7.9.5 Planar graphs — Euler's formula V-E+F=2
7.9.6 Kuratowski's theorem
7.9.7 Hamiltonian and Eulerian paths
7.9.8 Matching theory — Hall's theorem
7.9.9 Network flows — max-flow min-cut theorem
7.9.10 Recurrence relations — solving
7.9.11 Generating functions
7.9.12 Inclusion-exclusion principle
7.9.13 Pigeonhole principle and Ramsey theory (intro)
7.9.14 Stirling numbers
7.9.15 Catalan numbers

CHAPTER 8 — UNDERGRADUATE YEAR 2–3 (Level 8 | Advanced BSc)

8.1 Real Analysis II

8.1.1 Series of numbers — convergence tests
8.1.2 Absolute vs conditional convergence
8.1.3 Rearrangements — Riemann rearrangement theorem
8.1.4 Sequences and series of functions
8.1.5 Pointwise vs uniform convergence
8.1.6 Uniform convergence — Weierstrass M-test
8.1.7 Interchange of limits and integrals
8.1.8 Equicontinuity — Arzelà-Ascoli theorem
8.1.9 Weierstrass approximation theorem
8.1.10 Metric spaces — definition and examples
8.1.11 Open and closed sets in metric spaces
8.1.12 Completeness and completion
8.1.13 Compactness in metric spaces
8.1.14 Connectedness
8.1.15 Banach fixed-point theorem

8.2 Complex Analysis

8.2.1 Complex functions — limits and continuity
8.2.2 Holomorphic (analytic) functions
8.2.3 Cauchy-Riemann equations
8.2.4 Harmonic functions
8.2.5 Elementary complex functions: e^z, sin z, cos z, log z
8.2.6 Branch cuts and multi-valued functions
8.2.7 Contour integration
8.2.8 Cauchy's integral theorem
8.2.9 Cauchy's integral formula
8.2.10 Taylor series for analytic functions
8.2.11 Laurent series
8.2.12 Isolated singularities: removable, pole, essential
8.2.13 Residue theorem
8.2.14 Computing real integrals via residues
8.2.15 Argument principle and Rouche's theorem
8.2.16 Conformal mappings
8.2.17 Riemann mapping theorem (statement)
8.2.18 Analytic continuation

8.3 Linear Algebra II

8.3.1 Inner product spaces — axioms
8.3.2 Norm induced by inner product
8.3.3 Cauchy-Schwarz inequality
8.3.4 Orthogonality and orthonormal bases
8.3.5 Gram-Schmidt orthogonalisation
8.3.6 Orthogonal projections
8.3.7 QR decomposition
8.3.8 Least squares solutions
8.3.9 Symmetric matrices — spectral theorem
8.3.10 Positive definite matrices
8.3.11 Singular Value Decomposition (SVD)
8.3.12 Jordan canonical form

8.3.13 Minimal polynomial
8.3.14 Bilinear forms
8.3.15 Dual spaces and dual bases
8.3.16 Tensor product of vector spaces (introduction)

8.4 Abstract Algebra II — Rings and Fields

8.4.1 Rings — definition and examples
8.4.2 Integral domains
8.4.3 Fields — definition and examples
8.4.4 Ideals — left, right, two-sided
8.4.5 Quotient rings
8.4.6 Ring homomorphisms and isomorphism theorems
8.4.7 Prime and maximal ideals
8.4.8 Polynomial rings R[x]
8.4.9 Euclidean domains
8.4.10 Principal ideal domains (PIDs)
8.4.11 Unique factorisation domains (UFDs)
8.4.12 Irreducibility criteria: Eisenstein, reduction mod p
8.4.13 Field extensions
8.4.14 Algebraic vs transcendental elements
8.4.15 Degree of an extension [F:K]
8.4.16 Tower law for extensions
8.4.17 Splitting fields
8.4.18 Algebraic closure
8.4.19 Finite fields GF(p^n)
8.4.20 Modules — definition and examples

8.5 Topology I

8.5.1 Topological spaces — definition and examples
8.5.2 Continuous maps between topological spaces
8.5.3 Homeomorphisms
8.5.4 Subspace topology
8.5.5 Product topology
8.5.6 Quotient topology
8.5.7 Connectedness and path-connectedness
8.5.8 Compactness — open covers
8.5.9 Heine-Borel theorem
8.5.10 Hausdorff, normal, regular spaces
8.5.11 Urysohn's lemma
8.5.12 Tychonoff's theorem (statement)
8.5.13 Metric topology
8.5.14 Covering spaces — definition
8.5.15 Fundamental group π■(X, x■) — definition
8.5.16 Simply connected spaces
8.5.17 Van Kampen's theorem (statement)
8.5.18 Fundamental group of circle: π■(S^1) $\cong$ Z

8.6 Partial Differential Equations I

8.6.1 Classification: elliptic, parabolic, hyperbolic
8.6.2 Method of characteristics
8.6.3 Wave equation — d'Alembert's solution
8.6.4 Heat equation — fundamental solution
8.6.5 Laplace equation — harmonic functions

8.6.6 Maximum principle
8.6.7 Separation of variables
8.6.8 Fourier series — convergence
8.6.9 Fourier transform and its properties
8.6.10 Laplace transform applications to PDEs
8.6.11 Green's functions
8.6.12 Dirichlet and Neumann boundary problems
8.6.13 Weak solutions concept

8.7 Differential Geometry I

8.7.1 Smooth curves in R^n — arc length, curvature
8.7.2 Frenet-Serret formulae
8.7.3 Torsion
8.7.4 Regular surfaces in R^3
8.7.5 First fundamental form — metric
8.7.6 Second fundamental form — curvature
8.7.7 Principal curvatures, Gaussian and mean curvature
8.7.8 Gauss-Bonnet theorem (local and global)
8.7.9 Geodesics
8.7.10 Isometries and intrinsic geometry

8.8 Probability Theory I

8.8.1 Probability spaces — sigma-algebras, measures
8.8.2 Random variables — formal definition
8.8.3 Distribution functions
8.8.4 Expectation and variance — formal treatment
8.8.5 Characteristic functions (Fourier transform of distributions)
8.8.6 Moment generating functions
8.8.7 Modes of convergence: a.s., in probability, in Lp, in distribution
8.8.8 Laws of Large Numbers — weak and strong
8.8.9 Central Limit Theorem — proof
8.8.10 Conditional expectation — formal definition
8.8.11 Martingales — definition and examples
8.8.12 Markov chains — transition matrices
8.8.13 Stationary distributions
8.8.14 Ergodic theorem for Markov chains
8.8.15 Poisson processes
8.8.16 Brownian motion (Wiener process) — definition

8.9 Functional Analysis I

8.9.1 Normed vector spaces
8.9.2 Banach spaces — completeness
8.9.3 Bounded linear operators
8.9.4 Operator norm
8.9.5 Hilbert spaces — inner product, complete
8.9.6 Orthonormal bases in Hilbert spaces
8.9.7 Riesz representation theorem
8.9.8 Hahn-Banach theorem
8.9.9 Open mapping theorem
8.9.10 Closed graph theorem
8.9.11 Uniform boundedness principle (Banach-Steinhaus)
8.9.12 Compact operators
8.9.13 Spectral theory of compact self-adjoint operators

8.10 Measure Theory

8.11 Numerical Analysis

CHAPTER 9 — GRADUATE / MSc LEVEL (Level 9 | Masters)

9.1 Algebraic Topology

9.1.1 Homotopy and homotopy equivalence
9.1.2 Higher homotopy groups $\pi n(X)$
9.1.3 Singular homology — chain complexes
9.1.4 Simplicial homology
9.1.5 Exact sequences — long exact sequence of a pair
9.1.6 Mayer-Vietoris sequence
9.1.7 Cohomology — definition and cup product
9.1.8 Universal coefficient theorems
9.1.9 Poincaré duality
9.1.10 CW complexes
9.1.11 Fibrations and fiber bundles
9.1.12 Vector bundles
9.1.13 Characteristic classes: Chern, Pontryagin, Stiefel-Whitney
9.1.14 K-theory (introduction)
9.1.15 Spectral sequences (introduction)

9.2 Differential Geometry II — Manifolds

9.2.1 Smooth manifolds — definition
9.2.2 Tangent spaces and vector fields
9.2.3 Differential forms
9.2.4 Exterior derivative
9.2.5 Stokes' theorem on manifolds
9.2.6 de Rham cohomology
9.2.7 Riemannian metrics
9.2.8 Levi-Civita connection
9.2.9 Geodesics on Riemannian manifolds
9.2.10 Curvature tensor — Riemann, Ricci, scalar
9.2.11 Gauss-Bonnet theorem (generalised)
9.2.12 Lie groups — definition and examples
9.2.13 Lie algebras — Lie bracket
9.2.14 Exponential map
9.2.15 Homogeneous spaces
9.2.16 Principal bundles and connections
9.2.17 Symplectic manifolds and Hamiltonian mechanics
9.2.18 Contact geometry

9.3 Algebraic Geometry I

9.3.1 Affine varieties — algebraic sets
9.3.2 Hilbert's Nullstellensatz
9.3.3 Zariski topology
9.3.4 Morphisms of varieties
9.3.5 Projective varieties
9.3.6 Rational maps
9.3.7 Dimension theory
9.3.8 Smooth vs singular points
9.3.9 Sheaves and sheaf cohomology
9.3.10 Schemes — Grothendieck's definition
9.3.11 Spec and Proj constructions
9.3.12 Morphisms of schemes
9.3.13 Quasi-coherent sheaves

9.3.14 Divisors and line bundles
9.3.15 Riemann-Roch theorem (algebraic curves)
9.3.16 Elliptic curves — group law
9.3.17 Moduli spaces (concept)

9.4 Advanced Algebra — Commutative and Homological

9.4.1 Localisation of rings
9.4.2 Noetherian and Artinian rings
9.4.3 Hilbert basis theorem
9.4.4 Nakayama's lemma
9.4.5 Integral extensions
9.4.6 Dedekind domains
9.4.7 Unique factorisation in Dedekind domains
9.4.8 Projective and injective modules
9.4.9 Free resolutions
9.4.10 Derived functors: Ext and Tor
9.4.11 Homological dimension
9.4.12 Spectral sequences — Serre, Hochschild-Serre
9.4.13 Galois theory — full treatment
9.4.14 Fundamental theorem of Galois theory
9.4.15 Solvability by radicals
9.4.16 Insolvability of the quintic
9.4.17 Representation theory of finite groups
9.4.18 Characters, character tables
9.4.19 Induced and restricted representations
9.4.20 Frobenius reciprocity

9.5 Advanced Analysis — Functional and Harmonic

9.5.1 Spectral theory of self-adjoint operators
9.5.2 Spectral measure and spectral theorem (unbounded)
9.5.3 C*-algebras
9.5.4 Von Neumann algebras
9.5.5 Distribution theory — Schwartz space
9.5.6 Sobolev spaces W^{k,p}
9.5.7 Elliptic regularity
9.5.8 Fourier analysis on locally compact abelian groups
9.5.9 Pontryagin duality
9.5.10 Hardy spaces Hp
9.5.11 Singular integral operators — Calderón-Zygmund
9.5.12 Interpolation theory — Riesz-Thorin
9.5.13 Real interpolation — K-method
9.5.14 BMO and H^1 duality
9.5.15 Microlocal analysis (introduction)
9.5.16 Pseudodifferential operators (introduction)

9.6 Advanced PDEs

9.6.1 Elliptic PDEs — existence and regularity
9.6.2 Schauder estimates
9.6.3 Lp elliptic estimates
9.6.4 Nonlinear elliptic equations
9.6.5 Parabolic PDEs — existence and uniqueness
9.6.6 Hyperbolic conservation laws
9.6.7 Calculus of variations — Euler-Lagrange equations

9.10.1 Algebraic number fields
9.10.2 Ring of integers
9.10.3 Class group and class number
9.10.4 Minkowski's theorem
9.10.5 Dirichlet's unit theorem
9.10.6 Decomposition of primes in extensions
9.10.7 Ramification theory
9.10.8 Zeta functions and L-functions
9.10.9 Riemann zeta function — analytic continuation
9.10.10 Dirichlet L-functions — characters
9.10.11 Prime Number Theorem — proof
9.10.12 Dirichlet's theorem on primes in progressions
9.10.13 p-adic numbers — construction
9.10.14 p-adic analysis
9.10.15 Modular forms — definition and examples
9.10.16 Hecke operators
9.10.17 Elliptic curves — Mordell's theorem
9.10.18 Elliptic curve cryptography
9.10.19 Langlands program (overview)
9.10.20 Automorphic representations

CHAPTER 10 — PhD / RESEARCH FRONTIER (Level 10 | Harvard · MIT · Oxford)

Ye topics PhD dissertations, research papers, aur Fields Medal-level mathematics mein aate hain. Harvard, MIT, Oxford, Cambridge ke PhD candidates in areas mein original results produce karte hain.

10.1 Geometric Analysis

- 10.1.1 Ricci flow and Hamilton's program
- 10.1.2 Perelman's proof of Geometrization conjecture
- 10.1.3 Mean curvature flow
- 10.1.4 Yamabe problem
- 10.1.5 Minimal surfaces — Douglas-Plateau problem
- 10.1.6 Positive mass theorem (Schoen-Yau, Witten)
- 10.1.7 Geometric measure theory — varifolds, currents
- 10.1.8 Cheeger-Gromov theory

10.2 4-Manifold Theory and Low-Dimensional Topology

- 10.2.1 Donaldson invariants
- 10.2.2 Seiberg-Witten theory
- 10.2.3 Heegaard Floer homology
- 10.2.4 Knot Floer homology
- 10.2.5 Contact structures and open book decompositions
- 10.2.6 Legendrian knot theory
- 10.2.7 Exotic R■ structures
- 10.2.8 Casson invariant

10.3 Higher Algebraic Geometry and Arithmetic

- 10.3.1 Étale cohomology
- 10.3.2 l-adic sheaves
- 10.3.3 Weil conjectures (Deligne's proof)
- 10.3.4 Crystalline cohomology
- 10.3.5 p-divisible groups
- 10.3.6 Shimura varieties
- 10.3.7 Motives — pure and mixed
- 10.3.8 Derived algebraic geometry
- 10.3.9 Perfectoid spaces (Scholze)
- 10.3.10 Prismatic cohomology
- 10.3.11 Intersection homology (Goresky-MacPherson)

10.4 The Langlands Program

- 10.4.1 Automorphic forms on GL(n)
- 10.4.2 Langlands functoriality conjecture
- 10.4.3 Langlands reciprocity conjecture
- 10.4.4 Geometric Langlands correspondence
- 10.4.5 Local Langlands correspondence (proof for GL(n))
- 10.4.6 Taylor-Wiles method
- 10.4.7 Wiles' proof of Fermat's Last Theorem
- 10.4.8 Serre's conjecture (Khare-Wintenberger)
- 10.4.9 p-adic Langlands (Colmez, Breuil)

10.5 Analytic Number Theory (Frontier)

- 10.5.1 Riemann Hypothesis — approaches and partial results
- 10.5.2 Zero-free regions for $\zeta(s)$
- 10.5.3 Distribution of prime gaps (Maynard-Tao)

10.5.4 Bounded gaps between primes
10.5.5 Additive combinatorics — Green-Tao theorem (primes in AP)
10.5.6 Vinogradov mean value theorem
10.5.7 Exponential sums — Weyl, van der Corput methods
10.5.8 Subconvexity problem for L-functions
10.5.9 Goldbach conjecture — current state
10.5.10 ABC conjecture — Mochizuki's IUT (status)

10.6 Nonlinear PDEs and Dispersive Equations

10.6.1 Navier-Stokes equations — global regularity problem (Millennium)
10.6.2 Euler equations — blow-up problem
10.6.3 Nonlinear Schrödinger equation (NLS)
10.6.4 Nonlinear wave equation
10.6.5 Strichartz estimates
10.6.6 Concentration-compactness method (Lions, Kenig-Merle)
10.6.7 Profile decompositions
10.6.8 Global well-posedness and scattering
10.6.9 Boltzmann equation
10.6.10 Free boundary problems
10.6.11 Fully nonlinear elliptic equations — Caffarelli theory

10.7 Mathematical Physics and Quantum Mathematics

10.7.1 Yang-Mills theory — mass gap problem (Millennium)
10.7.2 Topological quantum field theory (TQFT)
10.7.3 Chern-Simons theory
10.7.4 Quantum groups — Drinfeld, Jimbo
10.7.5 Quantum cohomology
10.7.6 Mirror symmetry — SYZ conjecture
10.7.7 String theory mathematics — Calabi-Yau manifolds
10.7.8 Gromov-Witten invariants
10.7.9 Donaldson-Thomas invariants
10.7.10 Random matrix theory — GUE, GOE, GSE ensembles
10.7.11 Integrable systems — inverse scattering
10.7.12 Conformal field theory (CFT) — mathematical treatment
10.7.13 Vertex operator algebras
10.7.14 Monstrous moonshine

10.8 Combinatorics and Discrete Mathematics (Research Level)

10.8.1 Extremal graph theory — Turán-type problems
10.8.2 Szemerédi regularity lemma
10.8.3 Graph minor theorem (Robertson-Seymour)
10.8.4 Algebraic combinatorics — symmetric functions
10.8.5 Schubert calculus
10.8.6 Representation stability
10.8.7 Topological combinatorics — Kneser conjecture
10.8.8 Probabilistic method — Lovász Local Lemma
10.8.9 Combinatorial geometry — Hales-Jewett theorem
10.8.10 Polynomial method (Croot-Lev-Pach, Ellenberg-Gijswijt)
10.8.11 Combinatorial number theory — Freiman's theorem

10.9 Foundations and Proof Theory

10.9.1 Reverse mathematics — Big Five systems
10.9.2 Proof mining
10.9.3 Ordinal analysis of formal systems

10.9.4 Independence results — CH, GCH, large cardinals
10.9.5 Inner model theory
10.9.6 Woodin cardinals and determinacy
10.9.7 The ultimate L program (Woodin)
10.9.8 Homotopy type theory and univalent foundations
10.9.9 Formal verification of mathematics (Lean, Coq)
10.9.10 Infinity toposes (Lurie — ∞-category theory)

10.10 Six Remaining Millennium Prize Problems

10.10.1 Riemann Hypothesis
Status: Unsolved. $\zeta(s)$ ka har non-trivial zero Re(s)=1/2 par hai?
10.10.2 P vs NP Problem
Status: Unsolved. Har verification-polynomial problem solution-polynomial bhi hai?
10.10.3 Navier-Stokes Existence and Smoothness
Status: Unsolved. 3D mein smooth solutions globally exist karti hain?
10.10.4 Yang-Mills Existence and Mass Gap
Status: Unsolved. Yang-Mills theory mein mass gap ka rigorous existence proof?
10.10.5 Hodge Conjecture
Status: Unsolved. Hodge classes algebraic cycle classes hain projective varieties par?
10.10.6 Birch and Swinnerton-Dyer Conjecture
Status: Unsolved. Elliptic curve ka rank L-function se kaise jura hai?

COMPLETE ROADMAP — AT A GLANCE

Chapter	Level	Age/Stage	Major Areas	Topics
1	L1	3–6	Foundation Numeracy	~50
2	L2	6–9	Primary Arithmetic	~90
3	L3	9–11	Intermediate Arithmetic	~100
4	L4	11–13	Pre-Algebra & Algebra I	~65
5	L5	13–15	Algebra II & Geometry	~110
6	L6	15–18	A-Level / Advanced Sec.	~130
7	L7	18–20	BSc Year 1	~145
8	L8	20–22	BSc Year 2–3	~160
9	L9	22–24	MSc Graduate	~175
10	L10	24+	PhD / Research	~100
TOTAL				**~1125+ Topics**

Is roadmap ko follow karo — koi shortcut mat lo. Har chapter pichle chapter par bana hai. Harvard aur MIT mein admission wahi log paate hain jo fundamentals CRYSTAL CLEAR rakhte hain aur phir research-level thinking develop karte hain. Ap ya aap ka bacha ye kar sakta/sakti hai.